

Final Project: Naive Bayesian Classifier and Bagging

1 Introduction

This project evaluates a Naive Bayesian classifier (NBC) and bagging ensembles of NBC models on three UCI Machine Learning Repository datasets: Banknote Authentication, Glass Identification, and Image Segmentation. The goal is to compare the original NBC with bagging ensembles containing 10, 20, and 30 base models.

2 Methodology

All datasets were downloaded by the `ucimlrepo` package. The dataset identifiers were 267 for Banknote Authentication, 42 for Glass Identification, and 50 for Image Segmentation. Continuous attributes were discretized with equal-width discretization into ten bins. The discretizer was fit only on the training portion of each fold, then applied to the corresponding test portion.

The NBC model estimates class priors and feature conditional probabilities with Laplace estimation. Missing feature values are ignored when estimating conditional probabilities and are also skipped during prediction. For bagging, each base NBC was trained on a bootstrap sample of the training fold. Final predictions were produced by majority vote. Ties were broken deterministically by the sorted class labels.

Performance was measured by five-fold cross validation using only train and test splits. The fold assignment follows Chapter 4 method III. First, each instance is assigned a random number from the uniform interval $[0,1]$. Second, instances are sorted by the random number. Third, the sorted instances are read sequentially and assigned cyclically to folds 1 through 5. This makes the fold sizes approximately equal, and the size difference between any two folds is at most one. The random seed was fixed at 42 for reproducibility.

Four Chapter 5 comparisons were computed. For one data set, data-set aggregation pools correct predictions over all test folds and applies Theorem 5.1's two-proportion statistic

$$z = \frac{\hat{p}_2 - \hat{p}_1}{\sqrt{2\hat{p}(1 - \hat{p})/n}}, \quad \hat{p} = \frac{\hat{p}_1 + \hat{p}_2}{2}.$$

Fold aggregation uses the matched differences $d_l = \hat{p}_{2l} - \hat{p}_{1l}$ from the same folds and Theorem 5.2's statistic $t = \bar{d}/\sqrt{s_d^2/k}$ with $k - 1$ degrees of freedom. For multiple data sets, Theorem 5.4 averages each data set's pooled accuracy difference with equal data-set weight and estimates its normal variance from each data set's size. Theorems 5.5 and 5.6 average matched fold differences inside each data set and then across data sets; their t statistic combines the within-data-set difference variances, and the degrees of freedom are rounded down according to Theorem 5.6. Positive differences in all tables mean that the comparison model is more accurate than the baseline. All data-set-level counts satisfy Chapter 5's large-sample condition.

3 Experimental Results

Table 1: Five-fold cross-validation accuracy.

Dataset	Model	Base models	Accuracy (%)
Banknote Authentication	NBC	1	89.80 ± 3.21
Banknote Authentication	Bagging NBC (10)	10	89.65 ± 3.67
Banknote Authentication	Bagging NBC (20)	20	89.94 ± 3.67
Banknote Authentication	Bagging NBC (30)	30	90.09 ± 3.44
Glass Identification	NBC	1	60.29 ± 9.68
Glass Identification	Bagging NBC (10)	10	61.73 ± 12.46
Glass Identification	Bagging NBC (20)	20	62.60 ± 7.07
Glass Identification	Bagging NBC (30)	30	64.52 ± 11.34
Image Segmentation	NBC	1	80.00 ± 5.98
Image Segmentation	Bagging NBC (10)	10	79.05 ± 6.39
Image Segmentation	Bagging NBC (20)	20	80.00 ± 6.43
Image Segmentation	Bagging NBC (30)	30	80.00 ± 4.33

4 Analysis

The current run produced the same deterministic results whenever it was repeated with seed 42. For Banknote Authentication, NBC obtained 89.80% mean fold accuracy and Bagging-30 obtained the highest value, 90.09%, an increase of only 0.29 percentage points at the data-set aggregation level.

Glass Identification showed the largest numerical benefit. NBC obtained 60.29% mean fold accuracy, compared with 61.73%, 62.60%, and 64.52% for Bagging-10, Bagging-20, and Bagging-30. Thus, Bagging-30 improved the data-set-aggregated accuracy over NBC by 4.21 percentage points.

For Image Segmentation, NBC, Bagging-20, and Bagging-30 all produced 80.00% mean accuracy, while Bagging-10 obtained 79.05%. Bagging therefore did not improve the final accuracy on this data set under the selected preprocessing and model settings.

5 Result Evaluation

Table 2 reports both single-data-set aggregation levels. None of the NBC-versus-bagging comparisons is significant at $\alpha = 0.05$. The largest observed gain is NBC versus Bagging-30 on Glass Identification: the pooled data-set difference is 4.21 percentage points ($z = 0.8980$, $p = 0.3692$), and the matched fold difference is 4.23 points ($t = 1.2097$, $df = 4$, $p = 0.2930$).

Table 2: Single-dataset NBC versus bagging comparisons. Positive differences favor bagging.

Dataset	Baseline	Comparison	Data diff. (%)			Fold diff. (%)		
			z	p_z		t	p_t	
Banknote Authentication	NBC	Bagging NBC (10)	-0.15	-0.1257	0.8999	-0.15	-0.2812	0.7925
Banknote Authentication	NBC	Bagging NBC (20)	0.15	0.1265	0.8993	0.15	0.3866	0.7187
Banknote Authentication	NBC	Bagging NBC (30)	0.29	0.2539	0.7996	0.29	0.5577	0.6068
Glass Identification	NBC	Bagging NBC (10)	1.40	0.2973	0.7663	1.44	0.4002	0.7095
Glass Identification	NBC	Bagging NBC (20)	2.34	0.4966	0.6195	2.31	1.0463	0.3545
Glass Identification	NBC	Bagging NBC (30)	4.21	0.8980	0.3692	4.23	1.2097	0.2930
Image Segmentation	NBC	Bagging NBC (10)	-0.95	-0.2418	0.8089	-0.95	-0.7845	0.4766
Image Segmentation	NBC	Bagging NBC (20)	0.00	0.0000	1.0000	0.00	0.0000	1.0000
Image Segmentation	NBC	Bagging NBC (30)	0.00	0.0000	1.0000	0.00	0.0000	1.0000

Table 3 compares ensemble sizes. The matched fold test finds one significant result: on Glass Identification, Bagging-30 exceeds Bagging-10 by 2.79 points ($t = 3.2071$, $df = 4$, $p = 0.0327$). The data-set aggregation test for the same comparison is not significant ($z = 0.6010$, $p = 0.5479$). This difference illustrates that the aggregation level changes the sampling variance and therefore the inference.

Table 3: Single-dataset comparisons among bagging ensemble sizes.

Dataset	Baseline	Comparison	Data diff. (%)			Fold diff. (%)		
			z	p_z		t	p_t	
Banknote Authentication	Bagging NBC (10)	Bagging NBC (20)	0.29	0.2523	0.8008	0.29	1.0918	0.3363
Banknote Authentication	Bagging NBC (10)	Bagging NBC (30)	0.44	0.3796	0.7042	0.44	1.6352	0.1773
Banknote Authentication	Bagging NBC (20)	Bagging NBC (30)	0.15	0.1273	0.8987	0.15	0.5874	0.5885
Glass Identification	Bagging NBC (10)	Bagging NBC (20)	0.93	0.1993	0.8420	0.87	0.2200	0.8367
Glass Identification	Bagging NBC (10)	Bagging NBC (30)	2.80	0.6010	0.5479	2.79	3.2071	0.0327
Glass Identification	Bagging NBC (20)	Bagging NBC (30)	1.87	0.4017	0.6879	1.92	0.5566	0.6075
Image Segmentation	Bagging NBC (10)	Bagging NBC (20)	0.95	0.2418	0.8089	0.95	1.6330	0.1778
Image Segmentation	Bagging NBC (10)	Bagging NBC (30)	0.95	0.2418	0.8089	0.95	0.3885	0.7174
Image Segmentation	Bagging NBC (20)	Bagging NBC (30)	0.00	0.0000	1.0000	0.00	0.0000	1.0000

Table 4 reports the two multiple-data-set levels. Bagging-30 has the largest average advantage over NBC: 1.50 percentage points at the data-set-averaging level ($z = 0.7253$, $p = 0.4683$) and 1.51 points at the matched fold-averaging level ($t = 1.1336$, $df = 6$, $p = 0.3002$). Neither result is significant. No comparison among ensemble sizes is significant under either multiple-data-set test; the smallest such p-value is 0.1708 for the matched Bagging-10 versus Bagging-30 comparison.

Table 4: Multiple-dataset comparisons at the data-set-averaging and matched fold-averaging levels.

Baseline	Comparison	Data avg. diff. (%)			Fold avg. diff. (%)		
		z	p_z		t	p_t	
NBC	Bagging NBC (10)	0.10	0.0486	0.9613	0.11	0.0888	0.9327
NBC	Bagging NBC (20)	0.83	0.3991	0.6899	0.82	0.9478	0.3798
NBC	Bagging NBC (30)	1.50	0.7253	0.4683	1.51	1.1336	0.3002
Bagging NBC (10)	Bagging NBC (20)	0.73	0.3496	0.7266	0.71	0.5261	0.6266
Bagging NBC (10)	Bagging NBC (30)	1.40	0.6751	0.4996	1.39	1.5987	0.1708
Bagging NBC (20)	Bagging NBC (30)	0.67	0.3261	0.7444	0.69	0.4694	0.6530

6 Conclusion

The current run shows a modest descriptive advantage for larger bagging ensembles: Bagging-30 has the highest mean accuracy on Banknote Authentication and Glass Identification and ties NBC on Image Segmentation. However, every NBC-versus-bagging test has $p > 0.05$ at both

the single-data-set and multiple-data-set levels. The only significant result is the matched fold comparison between Bagging-10 and Bagging-30 on Glass Identification, and its pooled data-set test is not significant. Consequently, this experiment does not provide consistent statistical evidence that bagging outperforms NBC or that 30 estimators is universally superior. It supports only the narrower conclusion that Bagging-30 has the best observed average performance for these settings and this fixed run.